Porter Stemming Algorithm

Short Exam Notes — memorize the structure, not every rule.

Definition

Porter Stemmer is a rule-based stemming algorithm that reduces words to a common stem by applying a sequence of suffix-removal and replacement rules.

Overall Algorithm

1a	Plural endings	SSES → SS IES → I SS → SS S → ε
1b	Past / continuous	Remove ED or ING when the remaining part contains a vowel
1c	Y ending	Y → I when the stem contains a vowel
2	Longer suffixes	Replace/remove derivational suffixes
3	Further reduction	Reduce additional derivational suffixes
4	Suffix removal	Remove suffixes when the measure condition is satisfied
5	Final cleanup	Remove E / reduce final LL when conditions are satisfied

Step 1 — Easy Rules to Remember

1a: Plurals

```
SSES → SS     IES → I     SS → SS     S → ε
```

Examples: **caresses** → **caress**, **ponies** → **poni**, **cats** → **cat**.

1b: ED / ING

Remove **ED** or **ING** if the remaining part contains a vowel. Additional adjustments may follow, such as AT → ATE, BL → BLE, IZ → IZE, or reduction of an appropriate final double consonant.

1c: Y → I

Change Y to I when the stem contains a vowel. Example: **happy** → **happi**.

Step 2 — Representative Rules

Remember the pattern: **long suffix** → **shorter suffix**.

```
ATIONAL → ATE     TIONAL → TION     IZER → IZE     FUL → ε     NESS → ε
```

Example: **relational** → **relate**, **goodness** → **good**.

Step 3 — Representative Rules

```
ICATE → IC     ATIVE → ε     ALIZE → AL     ICITI → IC     ICAL → IC     FUL → ε     NESS → ε
```

Example: **electrical** → **electric**.

Step 4 — Common Suffixes

Common suffixes considered include: **AL, ANCE, ENCE, ER, IC, ABLE, IBLE, ANT, EMENT, MENT, ENT, ION, OU, ISM, ATE, ITI, OUS, IVE, IZE**.

These are removed only when the required **measure (m)** condition is satisfied.

Step 5 — Final Cleanup

```
Main rules: E → ε and LL → L, subject to the required conditions.
```

Memory Trick

1 = basic endings (S, ED, ING, Y) | **2–3 = bigger suffixes** | **4 = remaining suffixes** | **5 = final cleanup**

Exam Answer — 4 Lines

“Porter Stemmer is a rule-based stemming algorithm. It reduces words to stems by applying a sequence of suffix-removal and replacement rules. The algorithm is organized as Steps 1a, 1b, 1c, 2, 3, 4 and 5. It uses conditions

such as the measure of a word to avoid inappropriate suffix removal.”